Experiment 2: Installing a C Compiler in the Virtual Machine and Executing Simple Programs

Objective

To install a C compiler in a virtual machine created using VirtualBox and execute simple C programs.

Requirements

- A virtual machine running Linux or Windows
- A C compiler (GCC for Linux, MinGW for Windows)
- A text editor (Nano, Vim, or VS Code)

Steps to Install a C Compiler

Step 1: Open the Terminal or Command Prompt

- In Linux: Open the terminal.
- In Windows: Open the command prompt or PowerShell.

Step 2: Install a C Compiler

- **For Ubuntu/Debian**: sudo apt install gcc
- **For CentOS/RHEL**: sudo yum install gcc
- **For Fedora**: sudo dnf install gcc
- **For Windows (MinGW)**: Download and install MinGW from https://osdn.net/projects/mingw/

Step 3: Verify Installation

Run the following command to check if the compiler is installed:

```
gcc --version
```

Step 4: Write a Simple C Program

Create a file named hello.c using a text editor and add the following code:

```
#include <stdio.h>
int main() {
   printf("Hello, World!\n");
   return 0;
}
```

Step 5: Compile the C Program

Run the following command to compile the program:

gcc hello.c -o hello

Step 6: Run the Compiled Program

- **In Linux**: ./hello
- **In Windows**: hello.exe

Observations

- Check if the compiler installation was successful.
- Record any errors encountered and their solutions.
- Verify the output of the C program.

Result:

We have successfully installed a C compiler and executed a simple C program in a virtualized environment, reinforcing the understanding of software development in cloud computing.

Viva Questions

1. Why is a C compiler needed in a virtual machine?
2. How do you check if GCC is installed?
3. What are the differences between MinGW and GCC?
4. What command is used to compile a C program?
5. How do you execute a compiled C program in Linux and Windows?